



Quantum Advancement and Research Club (QARC)

IIT (ISM) Dhanbad



Q-Blueprint: A 4-Day Quantum Hackathon

Concetto'26 - The Annual Techno-Management Fest of IIT (ISM) Dhanbad

Problem Statement

Track: Pulse-Level Control, Transmon Physics, Drift Tracking, Adaptive Experiment Design

Start: 06 October 2026, 10:00 AM IST

Closed-Loop Auto-Calibration: Make the Qubit Tune Itself

Build a pulse-level simulator of a drifting three-level transmon, design an autonomous calibration agent that keeps single-qubit gates below a target error within a fixed shot budget, and quantify how the calibration policy propagates into the output quality of a variational quantum algorithm.

General Instructions

- **Thorough Review:** Read the complete problem statement, including the theoretical framework, before writing any code. Use the first part of Day 1 to understand the physical model and to derive the analytic limiting cases against which you will validate your simulator.
- **Timeline:** The hackathon commences on **06 October 2026 at 10:00 AM IST** and runs for four days. The phase-wise schedule below is an advisory guide; the final submission deadline is the only binding one.

Day	Phase	Expected status at the end of the day
1	Think	Theory understood; plant model implemented and validated against analytic limits.
2	Build	Calibration primitives and the end-to-end pipeline working on the static baseline.
3	Simulate	Policy comparison, Pareto sweeps and downstream-algorithm runs complete; first figures available.
4	Refine	Weaknesses resolved, critical analysis written, slides and live demo prepared.

- **Compute and Tools:** All tasks are designed to run on a laptop CPU with Python 3.10+ (NumPy, SciPy, Matplotlib, QuTiP, Qiskit, Qiskit Aer, Qiskit Dynamics). **No access to quantum hardware is required.** A full-day simulation should complete in minutes; if it does not, vectorise the propagator computation or cache intermediate results before scaling up.
- **Submission Format:** (i) clean, commented and reproducible Jupyter notebook(s) with fixed random seeds and a pinned list of dependencies, (ii) a written report (PDF) containing the analysis sections requested below, and (iii) a slide deck of at most 10 slides for the final presentation. The report must be self-contained and understandable without running the code.

- **Originality:** All code, analysis and written explanations must be your own work. Any external library, code snippet, paper or AI tool you use must be cited. During the final presentation you will be questioned on your code and results; you must be able to justify every component of your submission.
- **Scientific Integrity:** A negative or modest result that is correctly analysed scores higher than an overstated claim. Report distributions and uncertainty estimates, not only the best run.
- **Communication:** For queries regarding theory, specifications or debugging, contact the organising team through the official WhatsApp group.

Background and Motivation

A superconducting qubit is never “calibrated once and done”. Its transition frequency fluctuates by hundreds of kHz owing to two-level-system (TLS) defects and slow charge and flux noise; the effective drive gain varies with temperature and electronics; and the optimal pulse shape depends on both. A gate with 99.9% fidelity in the morning can degrade to 99% by noon. Processors with hundreds of qubits carry thousands of such parameters, so calibration must be automated, statistically efficient and inexpensive: every shot spent on calibration is a shot withheld from computation, and the device is unavailable to users while it is being calibrated.

This problem statement concerns the control layer between the device physics and the algorithm. You will model a driven three-level transmon, implement calibration primitives for amplitude, frequency and DRAG, make the frequency estimator efficient through Bayesian adaptive experiment design, embed the primitives in a policy that decides *when* to recalibrate, and finally measure how the policy changes the outcome of a quantum algorithm over a simulated 24-hour day. The central question is not whether a gate *can* be calibrated, but how little calibration suffices to keep an algorithm within tolerance.

Theoretical Framework

1. Driven Transmon with Frequency and Gain Drift

In the frame rotating at the nominal drive frequency ω_d , truncated to three levels, the driven transmon Hamiltonian is

$$H(t) = \Delta(t) a^\dagger a + \frac{\alpha}{2} a^\dagger a^\dagger a a + \frac{g(t)}{2} \left[\Omega_x(t) (a + a^\dagger) + \Omega_y(t) i(a^\dagger - a) \right], \quad (1)$$

where $\Delta(t) = \omega_q(t) - \omega_d$ is the drifting detuning, α is the anharmonicity, $\Omega_{x,y}$ are the programmed drive quadratures and $g(t)$ is the drifting drive gain. Within the qubit subspace, $a + a^\dagger \rightarrow \sigma_x$ and $i(a^\dagger - a) \rightarrow \sigma_y$.

Gaussian pulse and DRAG. A Gaussian pulse of duration t_g and width $\sigma = t_g/4$ has $\Omega_x(t) = A e^{-(t-t_0)^2/2\sigma^2}$, with A fixed by $\int \Omega_x dt = \theta$ for a rotation by angle θ . The DRAG correction adds a derivative term on the orthogonal quadrature,

$$\Omega_y(t) = -\beta \dot{\Omega}_x(t), \quad (2)$$

which suppresses the $|1\rangle \rightarrow |2\rangle$ transition. First-order theory gives $|\beta| \approx 1/|\alpha|$; the true optimum depends on pulse duration and amplitude and must be determined empirically. The sign of β_{opt} depends on convention and should be established from your own simulator.

2. Gate Error Including Leakage

Let U be the 3×3 propagator of a gate and $M = PUP$ its 2×2 block on the qubit subspace, which is non-unitary in the presence of leakage. For a target unitary V , the average gate fidelity of

the subspace operation is

$$\bar{F} = \frac{\text{Tr}(M_V M_V^\dagger) + |\text{Tr } M_V|^2}{d(d+1)}, \quad M_V = V^\dagger M, \quad d = 2, \quad (3)$$

and the gate error is $\epsilon = 1 - \bar{F}$ (Pedersen, Møller and Mølmer, 2007). Leakage out of the subspace is counted as error automatically. ϵ serves as the *oracle* score of your policy; the calibration agent itself never has access to it.

3. Calibration Primitives

Amplitude (error amplification). Suppose each nominal π pulse rotates by $\pi(1 + \eta)$. Preparing $|0\rangle$ and applying $R_X(\pi/2)$ followed by n such pulses gives, up to a convention-dependent sign,

$$\langle Z \rangle = -(-1)^n \sin(n\pi\eta), \quad (4)$$

which is linear in η for small η and amplified n -fold, so a few repetitions yield far better sensitivity than a single Rabi oscillation, until decoherence suppresses the signal.

Frequency (Ramsey). After $R_X(\pi/2) - \tau - R_X(\pi/2)$,

$$P(0 \mid \Delta, \tau) = \frac{1}{2} \left[1 + e^{-\tau/T_2} \cos(2\pi\Delta\tau + \varphi_0) \right]. \quad (5)$$

A single delay τ_0 is unambiguous only for $|\Delta| < 1/(2\tau_0)$ (aliasing), whereas long delays are more precise but alias sooner. A ladder of delays $\tau_k = 2^k \tau_0$ resolves both, as in iterative phase estimation. For N shots at delay τ with visibility $V = e^{-\tau/T_2}$, the Cramér–Rao bound is

$$\text{Var}(\hat{\Delta}) \geq \frac{1}{(2\pi\tau V)^2 N}, \quad (6)$$

which is optimised near $\tau \approx T_2$.

DRAG coefficient. Leakage is invisible to a binary readout, so indirect signatures are required, for example the growth of the apparent $P(1)$ after long trains of nominally identity pairs $[X_\pi X_{-\pi}]^n$ (leaked population reads as 1), or sequences whose error depends differently on β for the X and Y quadratures. The protocol is your choice, and you must justify it.

4. Adaptive (Bayesian) Experiment Design

Maintain a posterior $P(\Delta \mid \text{data})$ on a grid. After each batch of shots at delay τ_k , update it with the likelihood above, then select the next delay that minimises the expected posterior variance (or maximises the expected information gain). Compare against a fixed dense grid of delays at equal total shot count.

5. Drift Model: the Simulated Day

All teams use the same stochastic drift model so that results are comparable; seed **2026** defines the official trace.

- **Frequency:** $\Delta(t)/2\pi = x_f(t) + J(t)$, where x_f is an Ornstein–Uhlenbeck (OU) process (stationary standard deviation 150 kHz, correlation time 2 h) and $J(t)$ is random telegraph noise switching between 0 and +0.8 MHz with mean dwell time 4 h.
- **Gain:** $g(t) = 1 + 0.02 \sin(2\pi t/24 \text{ h} + \phi) + x_g(t)$, where x_g is an OU process (standard deviation 0.5%, correlation time 3 h) and ϕ is a phase fixed by the seed.

An OU process obeys $dx = -(x - \mu) dt/\tau_c + \sigma \sqrt{2/\tau_c} dW$ and can be sampled exactly on a uniform time grid of step δt via $x_{k+1} = \mu + (x_k - \mu) e^{-\delta t/\tau_c} + \sigma \sqrt{1 - e^{-2\delta t/\tau_c}} \xi_k$, with $\xi_k \sim \mathcal{N}(0, 1)$.

Simulation Specification

Parameter	Value
Anharmonicity	$\alpha/2\pi = -300$ MHz
Gate duration	$t_g = 20$ ns
Coherence times	$T_1 = 100 \mu\text{s}$, $T_2 = 80 \mu\text{s}$
Repetition time	$500 \mu\text{s}$ per shot
Levels retained	3 (leakage to $ 2\rangle$ included)
Readout	$P(1 0) = 1\%$, $P(0 1) = 3\%$; $ 2\rangle$ reads as 1
Simulated duration	24 h
Error threshold	$\epsilon_{\text{th}} = 10^{-3}$

Every shot advances the simulated clock by the repetition time. A calibration of N shots therefore costs $500 \mu\text{s} \times N$ of device time, during which the device keeps drifting and is unavailable to users. Use a fast piecewise-constant propagator (vectorised 3×3 matrix exponentials) and sample measurement outcomes from binomial distributions, so that a full simulated day runs in minutes.

Your Task

- Plant model (Day 1).** Implement the three-level transmon simulator with drift. Verify the static case: a Gaussian π pulse with $\int \Omega_x dt = \pi$, zero detuning and unit gain rotates the qubit while leaking measurably into $|2\rangle$. Plot $P_0(t)$, $P_1(t)$, $P_2(t)$ and report the leakage and the gate error ϵ from the fidelity formula above.
- Calibration primitives.** Implement three routines, each returning an estimate, an uncertainty and the number of shots consumed: (a) π -pulse amplitude (Rabi, refined by error amplification); (b) frequency (Ramsey, resolving aliasing and sign); (c) DRAG coefficient β . Demonstrate convergence to the known true values in a static test.
- Adaptive design.** Make the frequency calibration Bayesian. Show the posterior narrowing and the sequence of chosen delays τ . Plot estimation precision versus total shots for the adaptive scheme and for a fixed dense grid, and quantify the shot saving.
- Policy layer.** Implement and compare at least three policies over the simulated day: (P0) no recalibration after the initial calibration; (P1) recalibration on a fixed schedule, sweeping the period; (P2) a cheap health check every few minutes, with a full recalibration only when a test statistic exceeds a threshold that you justify. For every policy, record the full-day trace of the oracle gate error.
- Pareto analysis.** For each policy compute (A) the time-averaged oracle gate error and (B) the fraction of the day spent calibrating. Plot the Pareto frontier over the policy parameters, with error bars from at least 20 random drift seeds in addition to the official seed.
- Downstream algorithm.** Evaluate the effect of the calibration policy on a computation: a $p = 1$ QAOA for MaxCut on a 4-qubit ring. Implement every single-qubit gate using the instantaneous 2×2 block $M(t)$ obtained from your plant (rotation-amplitude error, detuning error and leakage included), applied through the trace-preserving map

$$\rho \mapsto M\rho M^\dagger + (1 - \text{Tr}[M\rho M^\dagger]) \frac{I}{2}.$$

Two-qubit R_{ZZ} gates are ideal. Plot the normalised cost $\langle C \rangle / C_{\text{max}}$ versus time of day for each policy.

Hardware–Algorithm Link

Your final figure must connect the calibration policy to algorithm quality, in the form: “Policy X spends $y\%$ of the day calibrating and delivers a $p = 1$ QAOA cost within $z\%$ of the ideal value for $w\%$ of the day.”

Deliverables

- **D1 Plant notebook:** transmon Hamiltonian, drift generators, gate-error oracle, static validation and population plots.
- **D2 Calibration primitives:** code and convergence plots for amplitude, frequency and DRAG; β_{opt} compared with $1/|\alpha|$ (sign convention stated).
- **D3 Adaptive versus fixed design:** precision-versus-shots plot, posterior-evolution plot, and a concise explanation of the information-theoretic origin of the saving.
- **D4 Policy comparison:** 24-hour traces of true versus estimated detuning and gain, oracle gate error versus time, and a Pareto plot with seed-to-seed error bars.
- **D5 Algorithm-level impact:** QAOA cost versus time of day under each policy.
- **D6 Memo (400 words):** “Which parameter drift hurts the algorithm most, and what would I change in the hardware or the control stack to need less calibration?” Cite at least two external sources.

Reporting standards. Label all axes with units, include legends and captions, state the number of seeds and shots behind every error bar, and specify whether uncertainties denote standard deviation, standard error or a confidence interval.

Bonus (outside the 100 marks). Replace the health check with a Kalman filter that tracks the OU state of the detuning from inexpensive measurements and triggers recalibration when the predicted gate error would exceed ϵ_{th} . Alternatively, remove the AC-Stark phase shift introduced by DRAG using a virtual- Z correction and demonstrate the resulting gain.

Learning Resources

Transmon Physics and DRAG

- Motzoi, F. *et al.* (2009). “Simple Pulses for Elimination of Leakage in Weakly Nonlinear Qubits.” *Phys. Rev. Lett.* 103, 110501. <https://arxiv.org/abs/0901.0534>
- Chen, Z. *et al.* (2016). “Measuring and Suppressing Quantum State Leakage in a Superconducting Qubit.” *Phys. Rev. Lett.* 116, 020501. <https://arxiv.org/abs/1509.05374>
- Krantz, P. *et al.* (2019). “A Quantum Engineer’s Guide to Superconducting Qubits.” *Appl. Phys. Rev.* 6, 021318. <https://arxiv.org/abs/1904.06560>
- Pedersen, L.H., Møller, N.M., Mølmer, K. (2007). “Fidelity of Quantum Operations.” *Phys. Lett. A* 367, 47. <https://arxiv.org/abs/quant-ph/0701138>

Automated Calibration and Adaptive Estimation

- Kelly, J. *et al.* (2018). “Physical Qubit Calibration on a Directed Acyclic Graph.” <https://arxiv.org/abs/1803.03226>
- Higgins, B.L. *et al.* (2007). “Entanglement-Free Heisenberg-Limited Phase Estimation.” *Nature* 450, 393. <https://arxiv.org/abs/0709.2996>

- Granade, C.E. *et al.* (2012). “Robust Online Hamiltonian Learning.” *New J. Phys.* 14, 103013. <https://arxiv.org/abs/1207.1655>

Tools

- Qiskit Dynamics: <https://qiskit-community.github.io/qiskit-dynamics/>
- Qiskit Experiments (calibration experiments): <https://qiskit-community.github.io/qiskit-experiments/>
- QuTiP documentation: <https://qutip.org/docs/latest/>

Evaluation

Submissions are scored out of 100 on five equally weighted criteria.

Criterion	What we look for	Marks
Conceptual depth	Correct understanding of the underlying physics; results explained through physical mechanisms rather than code behaviour.	/20
Implementation quality	Correct, modular, commented and reproducible code; no unexplained constants; self-consistent limiting cases and validations.	/20
Hardware–algorithm integration	The hardware-level result is carried through to a measurable effect on a computation, as the task requires.	/20
Analysis and critical thinking	Uncertainty quantification, comparison with theory, and honest treatment of limitations and failure cases.	/20
Communication and demo	Clear report, labelled plots (units, legends, captions) and a convincing final presentation with live demo.	/20

Bonus tasks are not part of the 100 marks; they are used to separate teams with similar scores. A poorly executed bonus does not cost marks, and a well-executed one can move a team to the top.